Fundamentals of Computer Vision (CS4059), Spring 2025

Programming Assignment-3

Automatic Georeferencing System using Local Feature based Image Matching

Nowadays Automatic Georeferencing System (AGS) is part of all modern UAV ground control systems (GCS) which can provide the geodetic location of any pixel in the live video feed of a UAV with a single mouse click. Some applications of AGS include: pinpointing the target locations in surveillance videos, artillery fire correction, UAV recovery and path planning. In this assignment you will implement an "Automatic Georeferencing System (AGS) using Local Feature based Image Matching" for an unmanned aerial vehicle (UAV). To complete the assignment, you need to complete following steps:

Step1: you are provided a reference/satellite image ***"reference.tif"*** and 10 UAV payload(camera) images from 'aerial1.png' to 'aerial10.png'. In this step you need to design an interface to load the reference and aerial images for viewing purposes.

Step2: User can select an aerial image and should be able to click any pixel/target on the image. You need to save the X,Y pixel coordinate of the clicked location.

Step3: Compute the keypoints and their descriptors for both selected aerial image and reference image. (for better efficiency you can also pre-compute the reference image keypoints and descriptors and store them in a file)

Step4: Compute the Homography matrix (H) using RANSAC to project any pixel of aerial image to reference image.

Step5: Using the H matrix project the clicked pixel location (step2) to reference image pixel location.

Step6: Using interpolation or simply rounding get the latitude and longitude of the projected pixel.

Step7: displays the clicked location coordinate to the user (it's better to display the coordinate on top of the image at the click location.

With all above steps user should be able to load any aerial image and click any pixel to get its coordinates

Important Points:

- Before extracting features you can manually rotate or align if required all aerial images to the reference image
- You can look at the demo codes of Week-5 to get help for local feature based image matching
- You need to give option to try at least two different detectors/descriptor methods (e.g., SIFT, HOG, SURF etc.)
- You also need to give a demo of your implementation before the due date